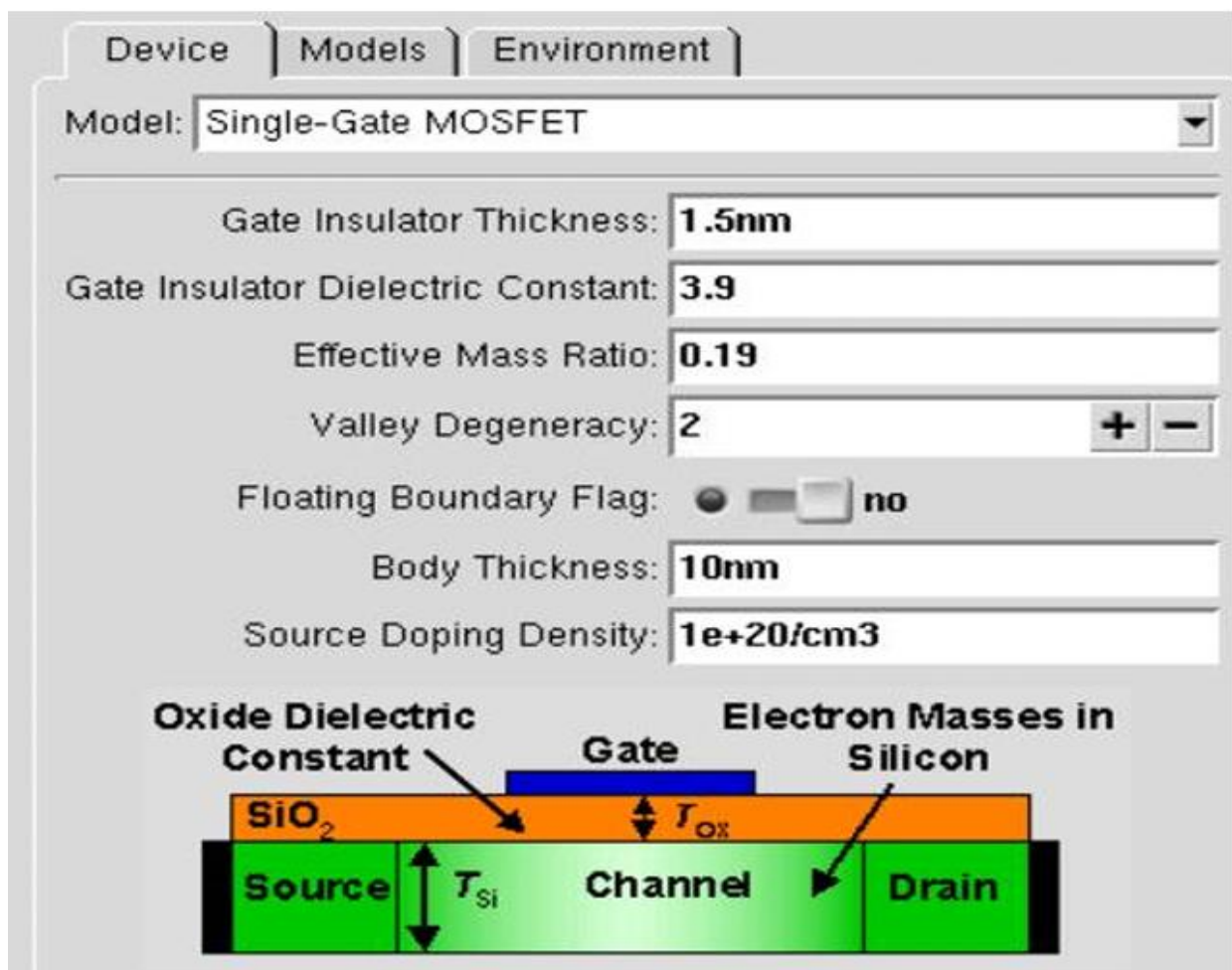


SLOT-C

ECA31032- NANO ELECTRONICS

Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Output:



Device Models Environment

Threshold Voltage: 0.32V

Gate Control Parameter: 0.88

Drain Control Parameter: 0.035

Series Resistance (ohm-um): 0

Device Models Environment

Ambient Temperature:  300K

Gate Voltage Sweep

Initial Gate Voltage:  0V

Final Gate Voltage:  1V

Gate Voltage Bias Points: 13  

Drain Voltage Sweep

Initial Drain Voltage:  0V

Final Drain Voltage:  1V

Drain Voltage Bias Points: 13  

Case 1: Increase in Gate Voltage

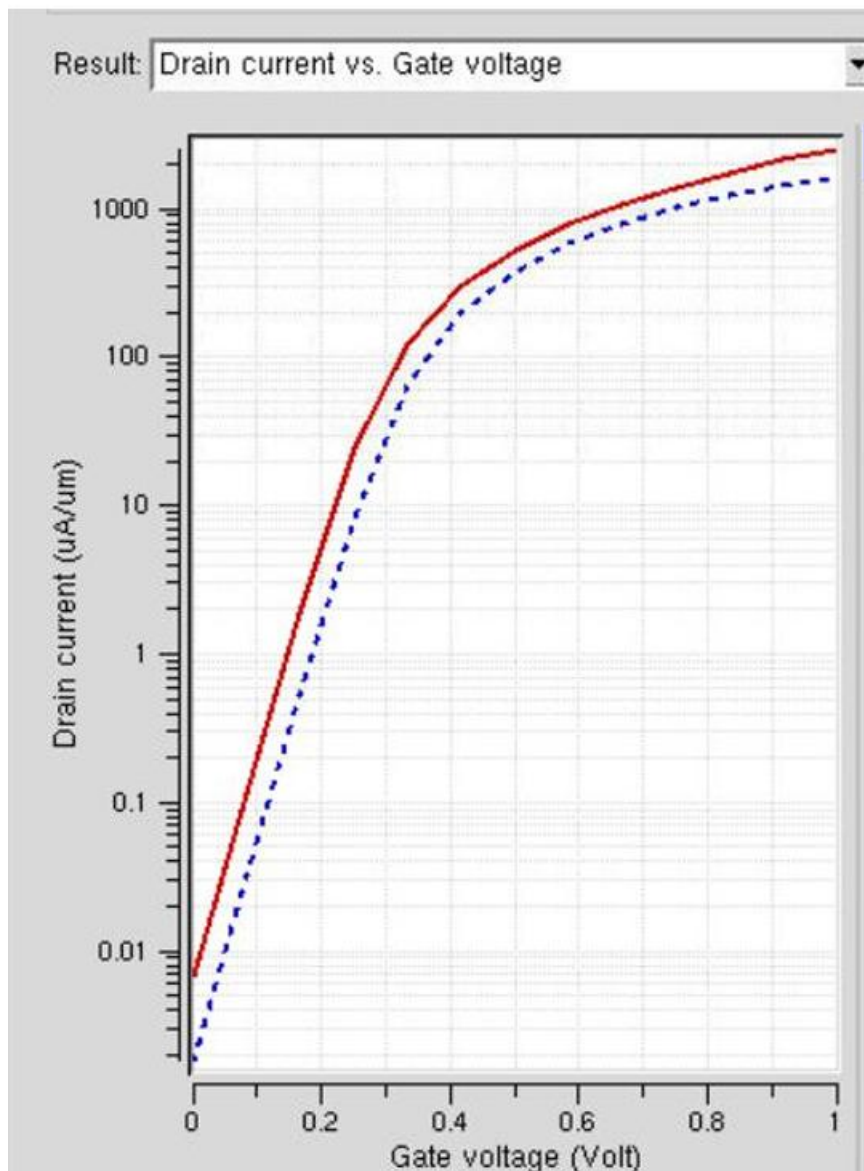
Explanation

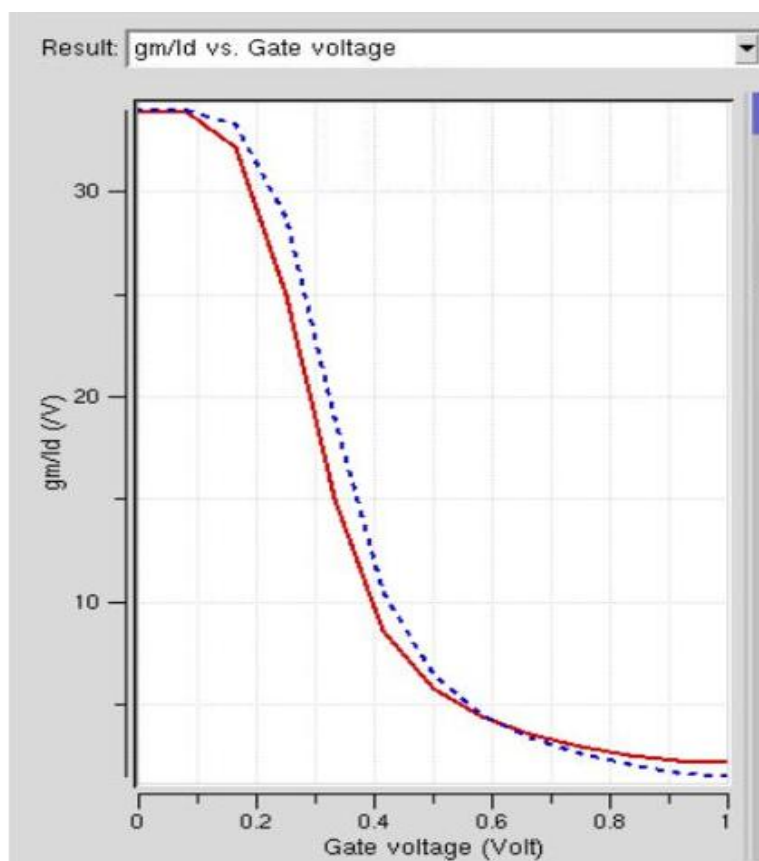
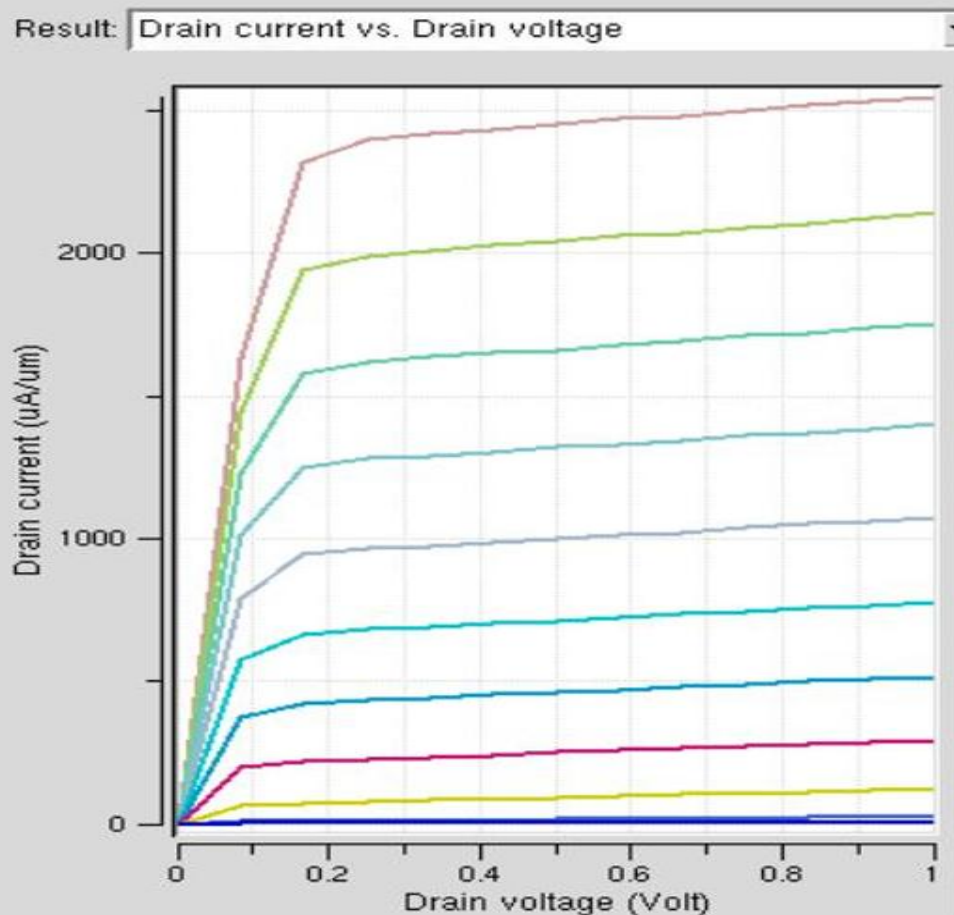
As the gate voltage (V_G) increases, more charge carriers are attracted to the channel, increasing its conductivity. As a result, the drain current (I_D) increases rapidly, and the MOSFET reaches the saturation region at a lower drain voltage.

Case 2: Variation in Drain Voltage

Explanation

At a fixed gate voltage (V_G), increasing the drain voltage (V_D) initially increases the drain current (I_D). Once the MOSFET enters the saturation region, the drain current becomes nearly constant even if V_D continues to increase. This indicates stable MOSFET operation in saturation.





Conclusion:

The analysis shows that the drain current (I_D) of the MOSFET is strongly influenced by both the gate voltage (V_G) and drain voltage (V_D). Increasing V_G enhances channel formation, resulting in a higher drain current, while increasing V_D causes the drain current to rise initially and then saturate. These characteristics confirm the normal operating behavior of the MOSFET and its suitability for amplification and switching applications.